Assignment 2: Data Analytics

Individual assignment | 100 points | **Deadline: 12 noon, Monday, 2 November**

Use the Gold-layer data prepared in Assignment 1 to answer four meaningful analytical questions. For each question, calculate appropriate measures, create a clear visualization, and explain the practical insight supported by the evidence.

1. Learning objectives

- Formulate meaningful questions that your available data can answer.
- Select and calculate suitable descriptive measures for the variables and question.
- Create clear visualizations and explain evidence-based insights.

Use your cleaned Assignment 1 dataset and Gold-layer star schema. You may refine questions from Assignment 1, but you must extend the earlier SQL work with suitable statistics, visualizations, and deeper interpretation. Revise any question that does not meet the category requirements below. Do not simply resubmit earlier results.

2. Choose four distinct analytical tasks

Choose exactly four of these seven categories: Trend; Comparison and Ranking; Part-to-Whole; Distribution; Relationship; Geographic; or Target/Deviation.

Comparison and Ranking count as one category. Each question must represent a different analytical purpose; changing only a filter, grouping field, or chart type does not create a new category. Select questions that your existing data can answer—no new dataset is required.

Before you begin, include a four-row planning table with: category, analytical question, relevant Gold tables and columns, intended measure, and a brief feasibility note explaining why the available data can answer the question.

3. Follow this workflow for each task

Use the five headings below for each of Tasks 1–4. Work iteratively: create an initial result, check the calculations and chart, then refine the final output.

Step 1. Ask one focused analytical question

Write a question that identifies what will be measured and compared, for whom or what, and over which period or scope. The question must be specific enough to answer with evidence.

Step 2. Identify variables and the analysis level

List the required columns, each variable's type (categorical, numeric, date/time, or Boolean), and its unit where relevant. State the unit of analysis (for example, one order line, customer, product, store, or month). Explain any filters, grouping, joins, and calculated measures.

Step 3. Calculate and show the evidence

Use SQL and/or Python to retrieve the required Gold-layer data. Show the result table and calculate only the descriptive statistics or basic measures needed to answer the question. Do not include every statistic simply because it is available.

Step 4. Create a suitable visualization

Create at least one final chart for the question using Seaborn, Matplotlib, or Plotly, and briefly justify why the chart fits the analytical purpose. Use a clear title, readable labels, units, meaningful ordering, honest scales, and consistent colors. Follow best practices when creating the charts and remove visual elements that do not help the reader understand the result.

Step 5. Interpretation and insight

Write approximately 100–150 words that: **(1) answer** the question directly, **(2) describe** the main pattern, **(3) explain** its practical meaning, and **(4) state** a reasonable implication or next step. Support the interpretation with at least two numerical results from your output. Do not claim causation or make predictions unless the analysis supports them.

5. Deliverables and submission

Submit to your course GitHub repository, following the Assignment 1 submission approach. Include:

- One executed Jupyter Notebook (.ipynb), or a clearly ordered set of notebooks, containing the four-row plan and all code, result tables, statistics, charts, and Markdown interpretations. All outputs must be saved and visible.
- A README.md file that acknowledges external code, adapted examples, and AI assistance according to the course academic-integrity policy. You are responsible for understanding, verifying, and explaining all submitted work.

Before submitting: restart the kernel and run all notebooks. Confirm that they finish without errors and regenerate all displayed tables and charts. Check that every question, measure, figure, and interpretation is consistent.

6. Grading rubric (100 points)

Each of Tasks 1–4 is worth 23 points (92 points total). Overall notebook quality, organization, and the README.md are assessed once for 8 points.

Criterion	Points	Full-credit evidence
Analytical question	3 × 4 = 12	Uses a distinct category and asks a focused, relevant, feasible question. Clearly identifies the variables, units, scope, and level of analysis.
Statistics and calculations	4 × 4 = 16	Uses appropriate, justified measures and correct Gold-based data. Joins, grain, filters, grouping, denominators, and calculations are accurate and checked.
Visualization quality	9 × 4 = 36	Selects a chart that fits the question and represents the results accurately. The chart has readable labels and units, an honest scale, meaningful ordering, consistent color, and a brief justification.
Interpretation and insight	7 × 4 = 28	Answers the question directly and supports the interpretation with at least two numerical results. Clearly explains the main observation, practical insight, and reasonable implication without unsupported claims.
Code quality and organization + README.md	8	Notebooks are clearly structured, readable, reproducible, and free of execution errors. Outputs are saved and visible; the README.md includes the required acknowledgments.
TOTAL	100	92 points for the four analyses + 8 points for the overall submission quality.

Partial credit: For each criterion, award 85–100% of the available points for complete, accurate, well-justified work; 70–84% for minor gaps; 50–69% for partial evidence or important errors; 1–49% for major weaknesses; and 0 for missing evidence. A missing task earns 0/23. Complexity alone does not earn extra credit.